

Objectives and Scope

Section 1 — Project Objective & Scope

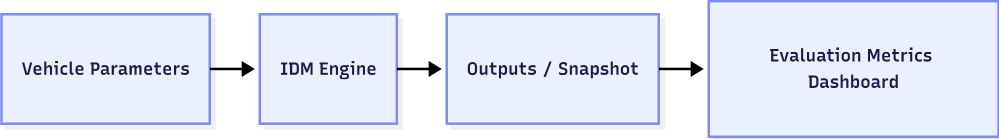
Category	Details
Project Goal	• Compare a fixed-time traffic signal and a roundabout at one intersection • Test both strategies under different traffic volumes • Use the same conditions for both strategies • Compare their performance using defined measurements
In Scope	• Single intersection • Motor vehicles • Fixed-time traffic signal • Roundabout • Realistic vehicle movement • Different traffic volumes • Performance measurements • Visual simulation • Repeatable simulations
Out of Scope	• Pedestrians and cyclists • Emergency vehicle priority • Multiple intersections • Real-world engineering validation • Declaring one strategy universally better • Additional control strategies

Success Criteria

#	Expected Result
1	• Both strategies run using the same user-defined conditions
2	• The same performance measurements are calculated for both strategies
3	• Results can be compared across different traffic volumes
4	• Simulation results are available as both visual and numerical outputs

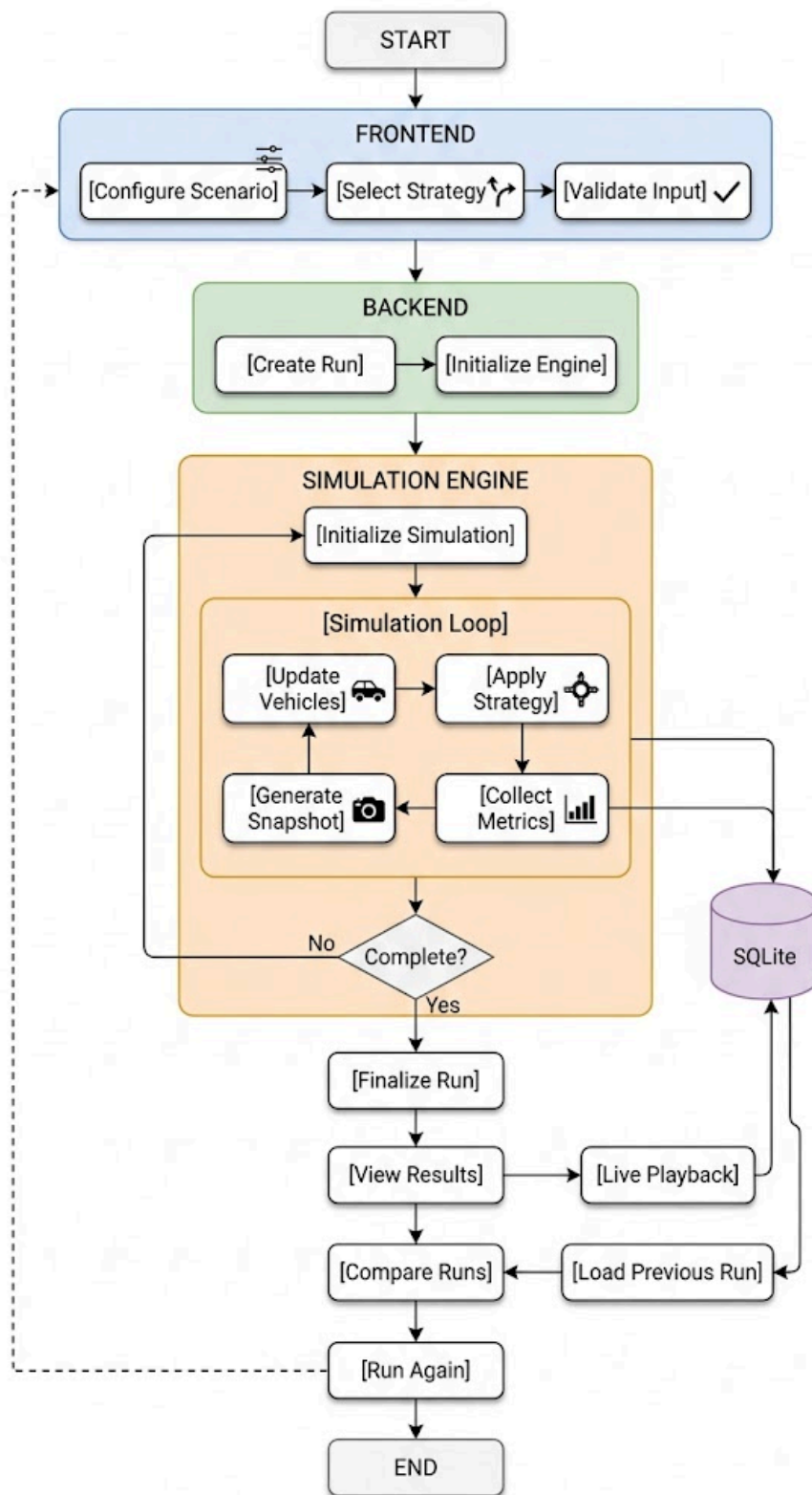
System Flow and Feature Breakdown

Section 3 — System Flow



Component Diagram

Step	Component	Purpose	Example Inputs / Data	Example Outputs / Data
1	Vehicle Parameters	• Defines the conditions used for the simulation	• Traffic volume • Number of lanes • Lane length • Vehicle speed	• Simulation conditions
2	IDM Engine (Intelligent Driver Model Engine)	• Simulates vehicle movement • Applies the selected traffic control strategy	• Vehicle parameters • Control strategy • Simulation duration	• Vehicle position • Vehicle speed • Vehicle state
3	Outputs / Snapshots	• Records simulation results during the run	• Vehicle position • Vehicle speed • Vehicle state • Simulation time	• Simulation data • Vehicle movement data
4	Evaluation Metrics Dashboard	• Displays the simulation results • Compares both strategies	• Simulation results • Performance measurements	• Numerical results • Visual results • Strategy comparison



Flow Diagram of the entire Application

Section 4 — Key Features

#	Feature	What It Does
1	Vehicle Simulation	• Simulates realistic vehicle movement
2	Traffic Control	• Simulates fixed-time traffic signals • Simulates roundabouts
3	Performance Measurement	• Measures vehicle and intersection performance
4	Strategy Comparison	• Compares both strategies under the same conditions
5	Traffic Volume Analysis	• Tests different traffic volumes
6	Simulation History	• Saves previous simulation results for later viewing and comparison
7	Visual Simulation	• Shows vehicle movement during simulation
8	User Parameters	• Allows users to change simulation conditions
9	Results Dashboard	• Displays measurements and comparisons

Vehicle Motion Model (Intelligent Dri...

Section 5 — Vehicle Motion Model (Intelligent Driver Model (IDM))

Why Intelligent Driver Model (IDM)

Point	Details
1	• Widely used car-following model
2	• Produces smooth acceleration and braking
3	• Uses a small number of understandable parameters
4	• Works independently of the traffic control strategy
5	• Keeps vehicle behavior consistent for both the fixed-time traffic signal and roundabout

$$\dot{v} = a \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right]$$

$$s^*(v, \Delta v) = s_0 + vT + \frac{v \Delta v}{2\sqrt{ab}}$$

Parameters			
Symbol	Name	Typical value	Meaning
v_0	Desired speed	13.9 m/s (~50 km/h)	Free-flow speed with no vehicle ahead
a	Max acceleration	1.0-1.5 m/s ²	How eagerly the vehicle accelerates toward v_0
b	Comfortable braking decel.	1.5-2.0 m/s ²	Normal (non-emergency) braking rate
δ	Acceleration exponent	4	Sharpness of accel taper as $v \rightarrow v_0$
s_0	Minimum jam gap	~2 m	Gap maintained at full standstill
T	Safe time headway	~1.5 s	Desired following gap, scales with speed
s	Actual gap	computed/step	Bumper-to-bumper distance to vehicle ahead
Δv	Approach rate	computed/step	$v - v_{lead}$; positive = closing in

Governing Equation

Part	Purpose
First part	• Calculates the vehicle's tendency to accelerate on an open road
Second part	• Calculates the desired distance from the vehicle ahead
Overall equation	• Calculates the vehicle's acceleration or deceleration at a given time

Application to Intersection Control

Controller Decision	IDM Input	Vehicle Behavior
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May proceed = No	- Distance to the stopping or yielding point - Speed difference between the vehicle and a stationary reference	Vehicle slows down or stops at the required point
May proceed = Yes	- Distance to the actual vehicle ahead - Speed difference with the vehicle ahead	Vehicle follows the vehicle ahead normally
No vehicle ahead	Open-road conditions	Vehicle moves toward its desired speed

Vehicle Control Flow

Step	Action
1	Controller determines whether the vehicle may proceed
2	If No , the vehicle treats the stopping or yielding point as a stationary reference
3	If Yes , the vehicle follows the actual vehicle ahead
4	IDM calculates acceleration or deceleration
5	Vehicle movement is updated

Edge Cases

#	Case	Handling
1	Zero or negative distance	- Use a small positive distance - Prevent calculation errors
2	Vehicle stopped at stopping point	- Acceleration should remain approximately zero - Prevent repeated starting and stopping
3	Vehicle ahead is moving away	- Keep the speed difference in the calculation - Do not force the value to zero - Preserve normal vehicle-following behavior

Control Strategy Logic

Section 6 — Traffic Control Strategies

Feature	Fixed-Time Traffic Signal	Roundabout
Control Method	• Uses fixed signal timings	• Uses available gaps in traffic
Vehicle Movement	• Vehicles move when the signal allows them to proceed	• Vehicles enter when a safe gap is available
Traffic Response	• Timing does not change with traffic volume	• Decisions respond to current traffic
Low Traffic	• Vehicles may still wait for the signal	• Vehicles can proceed when a safe gap is available
Main Difference	• Uses fixed timing	• Uses current traffic conditions

Evaluation Metrics Short

Section 7 — Evaluation Metrics

Category	Metric	What It Measures
Efficiency	Average Wait Time	• How long vehicles wait
	Throughput	• How many vehicles pass through
	Queue Length	• How many vehicles are waiting
Traffic Flow	Stop Count	• How often vehicles stop
	Speed Variance	• How much vehicle speeds vary
	Travel Time Reliability	• How consistent travel times are
	Average Travel Speed	• How quickly vehicles travel through the intersection
System Performance	Idle Opportunity Loss	• Time vehicles wait when they could safely move
	Critical Saturation Volume	• Traffic level where performance starts to decline
	Intersection Utilization	• How much of the available intersection capacity is being used
Fairness & Stability	Directional Fairness Index	• How evenly waiting time is distributed
	Queue Stability Index	• How much queues change over time

	Congestion Recovery Time	• How quickly congestion reduces
Physical Constraints	Space / Footprint Consumed	• How much physical space the intersection requires

Execution Plan

Section 8 — Execution Plan for 2 Months

Week	Dates	Expected Output / Sprint Goal	Viraj – Simulation / Data	Khushi – Frontend / Analytics
W1	Aug 3 – Aug 9	V0.1 – Single Vehicle • Vehicle movement • Basic simulation	• Set up simulation • Implement vehicle movement	• Set up user interface • Display simulation
W2	Aug 10 – Aug 16	V0.2 – Multiple Vehicles • Vehicle interaction • Vehicle following	• Extend vehicle movement • Implement vehicle following	• Display multiple vehicles • Display vehicle movement
W3	Aug 17 – Aug 23	V0.3 – Traffic Control Strategies • Fixed-time traffic signal • Roundabout	• Implement both strategies • Apply to same simulation	• Display both strategies • Display active strategy
W4	Aug 24 – Aug 30	V0.4 – Simulation Outputs • Snapshots • Vehicle data • Simulation results	• Generate snapshots • Record vehicle states and movement	• Display vehicle states • Add simulation playback
W5	Aug 31 – Sep 6	V0.5 – Performance Metrics • Defined metrics • Results for both strategies	• Calculate metrics • Apply consistently • Validate results	• Display metrics • Display metrics with playback
W6	Sep 7 – Sep 13	V0.6 – Strategy Comparison • Direct comparison • Same test conditions	• Run identical scenarios • Collect comparison results • Ensure repeatability	• Add side-by-side comparison • Add tables and charts
W7	Sep 14 – Sep 20	V0.7 – Traffic Volume Analysis • Multiple traffic volumes • Volume-based comparison	• Run volume sweeps • Store and compare results	• Add simulation history • Add volume comparison charts

W8	Sep 21 – Sep 27	V1.0 – Final Validated Study - Validated results - Complete user interface - Final output	- Test inputs - Validate results - Check repeatability - Prepare final data	- Finalize dashboard - Finalize comparisons - Prepare final results
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Proof of Concepts

Section 9 — Proof of Concepts

#	Validation Check	What We Check	Expected Result
1	Vehicle Movement	• Vehicles move realistically • Vehicles maintain safe spacing	• Smooth movement • No collisions
2	Control Strategies	• Both strategies work under the same conditions	• Both produce valid results
3	Performance Measurements	• Measurements are calculated consistently	• Same calculation method for both
4	Traffic Volume	• Results change as traffic increases	• Meaningful changes in results
5	Visual Simulation	• Vehicle movement is displayed correctly	• Clear and synchronized movement
6	User Inputs	• Changed inputs affect the simulation	• New inputs produce new results

Risks & Assumptions, Future Scope

Section 10 — Risks

#	Risk	Mitigation
1	Vehicle behavior may become unrealistic at extreme traffic volumes	• Test realistic traffic ranges • Review unusual results
2	Performance measurements may not be consistent	• Use the same definitions for both strategies • Validate measurements
3	Development may take longer than planned	• Prioritize core features • Remove lower-priority features if needed
4	Visual results may not match numerical results	• Compare visual and numerical results during validation

Section 11 — Future Scope

#	Deferred Feature	Purpose
1	Run History / Persistence	• View previous configurations and results
2	Hybrid Control Model	• Add a third control strategy
3	Natural Language Parameter Input	• Set simulation parameters using natural language
4	User-Weighted Recommendation Scoring	• Provide a recommendation based on selected metric weights
5	Additional Metrics	• Add emissions, fuel consumption, and safety-related measurements
6	Advanced Traffic Conditions	• Test uneven traffic volumes and extreme congestion

References

References

Vehicle Motion Model

#	Reference	Used For
1	Treiber, M., Hennecke, A., & Helbing, D. (2000). <i>Congested traffic states in empirical observations and microscopic simulations</i> . Physical Review E, 62(2), 1805–1824. Original paper — American Physical Society	• Intelligent Driver Model (IDM) • Section 5
	DOI: 10.1103/PhysRevE.62.1805 DOI link	

Roundabout Control Logic

#	Reference	Used For
2	Transportation Research Board / National Academies. <i>Highway Capacity Manual (HCM)</i> . Highway Capacity Manual — Transportation Research Board	• Roundabout capacity and operations • Gap-acceptance concepts • Section 6.2
3	Tanner, J. C. (1962). <i>A Theoretical Analysis of Delays at an Uncontrolled Intersection</i> . Biometrika, 49(1/2), 163–170. Original paper — JSTOR	• Gap-acceptance theory • Section 6.2
4	Federal Highway Administration (FHWA). <i>Roundabouts: An Informational Guide — Second Edition</i> . FHWA Roundabouts Guide	• Roundabout operation • Capacity • Vehicle interactions • Roundabout design and evaluation
5	National Academies. <i>Roundabouts: An Informational Guide — Second Edition (NCHRP Report 672)</i> . National Academies — NCHRP Report 672	• Roundabout operation • Planning and evaluation • Supporting reference for Section 6.2

Traffic Signal Control

#	Reference	Used For
6	Federal Highway Administration (FHWA). <i>Traffic Signal Timing Manual</i> . FHWA Traffic Signal Timing Manual	• Signal timing • Signal phases • Signal control concepts • Section 6.1
7	Federal Highway Administration (FHWA). <i>Traffic Signal Timing Manual</i> — PDF. Download PDF	• Full reference document • Signal timing and control practices
8	FHWA. <i>Traffic Signal Timing & Operations Strategies</i> . FHWA Signal Timing & Operations	• Signal timing concepts • Performance-based signal operations

Evaluation Metrics

#	Reference	Used For
9	Federal Highway Administration (FHWA). <i>Traffic Signal Timing Manual</i> . FHWA Traffic Signal Timing Manual	• Average Wait Time • Stop Count • Signal performance measures
10	Federal Highway Administration (FHWA) / Highway Capacity Manual (HCM).	• Throughput • Queue Length • Intersection performance
11	FHWA. <i>Does Travel Time Reliability Matter?</i> FHWA Travel Time Reliability Primer	• Travel Time Reliability • Travel time variability • Reliability measures
12	FHWA. <i>Travel Time Reliability: Making It There On Time, All The Time</i> . FHWA Travel Time Reliability Report	• Travel Time Reliability • 90th / 95th percentile travel time • Reliability measurement
13	FHWA. <i>Analyzing Travel Time Reliability: Data and Analysis Methods</i> . FHWA Travel Time Reliability Analysis	• Travel time variability • Standard deviation • Percentile-based reliability measures

Data & Simulation History Storage

Section 7 — Data & Simulation History Storage

Aspect	Implementation
Storage	• Local SQLite database on the user's machine • No separate database server required
Why	• Simple & lightweight — no separate database server or complex setup. • Local & portable — entire database can be stored as a single file with the application. • Sufficient for project scale — easily handles simulation runs, snapshots, and metrics. • Easy history retrieval — supports storing and querying past runs without rerunning simulations.
Run Information	• Unique Run ID • Timestamp • Signal / Roundabout strategy • Run status
Scenario Data	• Traffic volume • Lane count • Geometry • Vehicle parameters • Random seed
Snapshots	• Simulation timestamp • Vehicle position • Speed • Vehicle state
Metrics	• Average wait time • Average travel speed • Stop count • Throughput • Queue length
Data Flow	Run → Snapshots → Metrics → SQLite → Dashboard
History Access	• Select previous Run ID • Load stored snapshots/metrics • View or compare results without rerunning

Purpose	• Persistent history • Faster result access • Reproducible runs • Historical comparison
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Evaluation Metrics Detailed

Section 7 — Evaluation Metrics

Measurement Conditions

Parameter	Definition
Comparison	• Fixed-time traffic signal vs. roundabout
Traffic Demand	• Same traffic demand for both strategies
Road Geometry	• Same road geometry for both strategies
Vehicle Model	• Same vehicle model for both strategies
Simulation Duration	• Same simulation duration for both strategies
Metric Definitions	• Same definitions and calculation rules for both strategies

Metric Categories

Category	Measures
Operational Efficiency	• Intersection efficiency

	• Vehicle processing
Traffic Flow Quality	• Vehicle movement • Travel consistency
System Performance	• Capacity usage • Traffic service
Fairness & Stability	• Delay balance • Queue stability • Congestion recovery
Physical Constraints	• Required intersection space

7.1 — Operational Efficiency

Metric	Definition / Calculation	Rule
Average Wait Time	• Average time vehicles spend stopped or crawling • AWT = Total vehicle wait time / Completed vehicles	• Waiting: speed < 1 m/s
Throughput	• Vehicles that completely exit the intersection per hour	• Count only vehicles crossing the defined exit boundary

	• Throughput = Vehicles cleared / Measurement time × 3600	
Queue Length	• Number of waiting vehicles at each time step • Report Average, Maximum, and P95 queue length	• Queue: speed < 1 m/s within the influence zone

7.2 — Traffic Flow Quality

Metric	Definition / Calculation	Rule
Stop Count	• Number of separate stops per vehicle • Average Stops = Total stops / Completed vehicles	• Stop: speed changes from $\geq 1 \text{ m/s}$ to $< 1 \text{ m/s}$ • Continuous stopping counts as one stop
Speed Variance	• Measures variation in vehicle speed • $\text{Var}(v) = \Sigma(v_i - \bar{v})^2 / (n-1)$	• Calculate within the influence zone
Travel Time Reliability	• Measures consistency of travel time • Report Mean, Standard Deviation (SD), and P95	• Travel Time = Exit time – Entry time
Average Travel Speed	• Average speed of completed vehicles through the intersection • ATS = Total travel distance / Total travel time	• Use the same entry and exit boundaries • Include only completed vehicles

7.3 — System Performance

Metric	Definition / Calculation	Rule
Idle Opportunity Loss	• Time vehicles wait when a safe opportunity to proceed is available	• Waiting vehicle • Safe opportunity available • No vehicle being served
Critical Saturation Volume	• Traffic volume at which performance starts to continuously degrade	• Based on predefined delay, queue, and throughput criteria
Intersection Utilization	• Percentage of active-demand time during which vehicles are being served • Utilization = Service time / Active-demand time × 100	• Exclude periods with no approaching or waiting vehicles

7.4 — Fairness & Stability

Metric	Definition / Calculation	Rule
Directional Fairness Index	• Measures how evenly delay is distributed between approaches • DFI = Standard Deviation (SD) of approach average wait times	• Lower value = more balanced delay
Queue Stability Index	• Measures queue variation over time	• Lower value = more stable queue

	• $QSI = SD(\text{queue length}) / \text{Mean}(\text{queue length})$	
Congestion Recovery Time	• Time required for the queue to return to normal after heavy congestion • Recovery Time = Normal-condition time – Congestion-end time	• Normal-condition threshold fixed before testing

7.5 — Physical Constraints

Metric	Definition / Calculation	Rule
Space / Footprint Consumed	• Physical area required by the intersection geometry • Examples: $A = L \times W$, $A = \pi r^2$	• Independent of traffic volume

Common Measurement Rules

Parameter	Definition
Influence Zone	• Fixed road area around the intersection used for measurements
Crawl-Speed Threshold	• 1 m/s initially• Final value fixed before main testing
Completed Vehicle	• Vehicle that enters and fully exits the measurement zone
Warm-up Period	• Initial simulation period excluded from metric calculations

Measurement Period	• Fixed period after warm-up used for final calculations
Random Seed	• Same seed and traffic demand used for corresponding signal and roundabout runs